

STAINLESS STEEL WELDED TUBINGS

WELDED STAINLESS STEEL TUBING

MECHANICAL PROPERTIES

Type	Round and square			Round			
	Tension test			Bending	Flattering		Flaring
	Yield strength kg/mm ²	Tensile strength kg/mm ²	Elongation (longitudinal) %		No.1	No.2	
AISI 430	> 25	> 42	> 20	180°4D	< ² / ₃ D	< ¹ / ₂ D	1.14D
AISI 304	> 21	> 53	> 35	180°3D	< ² / ₃ D	< ¹ / ₂ D	1.2D
AISI 316	> 21	> 53	> 35	180°3D	< ² / ₃ D	< ¹ / ₂ D	1.2D

Notes:

- (1) Yield strength test shall not be conducted unless requested. Other mechanical property values are available on request.
- (2) For the tension test of round tubing, test piece No. 11 or 12 specified in JIS Z 2201 (Tension Test Pieces for Materials) shall be used. Test piece No. 12 shall be taken from an unwelded portion.
- (3) For the tension test of square tubing, test tubing, test piece No. 5 specified in JIS Z 2201 shall be used. However, for a side length of less than 25 mm, the test shall be conducted on a test piece with a width equivalent to its side length. The test piece shall be taken from an unwelded portion.
- (4) Unless otherwise specified, the values of No. 1 shall be used.

TYPES OF MANUFACTURES

Surface Finish			
	Finish	Nisshin Code	Description
Round tubing	Unpolished	S2	As welded
	120 GRIT	S 240	Polished by #120 belt
	120 GRIT	SB 240	Buffed S 240
	180 GRIT	SB 400	Buffed after polishing by #180 belt
	180 GRIT	GL	Hair-line finished after polishing by #180 belt
	320 GRIT	SB 600	Buffed after polishing by #320 belt
	400 GRIT	BL 400	Buffed after polishing by #400 belt
	MIRROR	MR	Buffed after polishing by #600 belt
Square & rectangular tubing	Unpolished	S2	As welded
	180 GRIT	HL180	Hair-line finished after polishing by #180 belt

CHEMICAL COMPOSITION

Type	Chemical Composition (%)							
	C	Si	Mn	P	S	Ni	Cr	Mo
AISI 430	<0.12	<1.00	<1.00	<0.04	<0.03	--	16 ~18	--
AISI 304	<0.08	<1.00	<2.00	<0.04	<0.03	8~11	18~20	--
AISI 316	<0.08	<1.00	<2.00	<0.04	<0.03	11~14	16 ~18	200 ~300